Customer Shopping Behavior Analysis

Uncovering insights from transactional data to guide strategic business decisions.



Project Overview & Data Summary

Project Goal

Analyze 3,900 purchases to identify patterns, segments, product preferences, and subscription behavior.

This analysis aims to inform key business strategies.

Dataset Snapshot

- Rows: 3900, Columns: 18
- Key Features: Demographics, Purchase Details, Shopping Behavior.
- Missing Values: 37 in Review Rating.



Made with GAMMA

EDA using Python: Data Preparation

01

Data Loading & Initial Understanding

Import dataset using pandas; use `df.head()`, `df.info()`, `df.describe()`, and `df.isnull().sum()` for initial checks.

02

Handling Missing Values

Missing values in 'Review Rating' were filled using median imputation based on product category.

03

Column Name Standardization

Converted column names to lowercase and replaced spaces with underscores for consistency.

04

Feature Engineering

Created 'age_group' using quartile segmentation and 'purchase_frequency_days' from textual frequency.

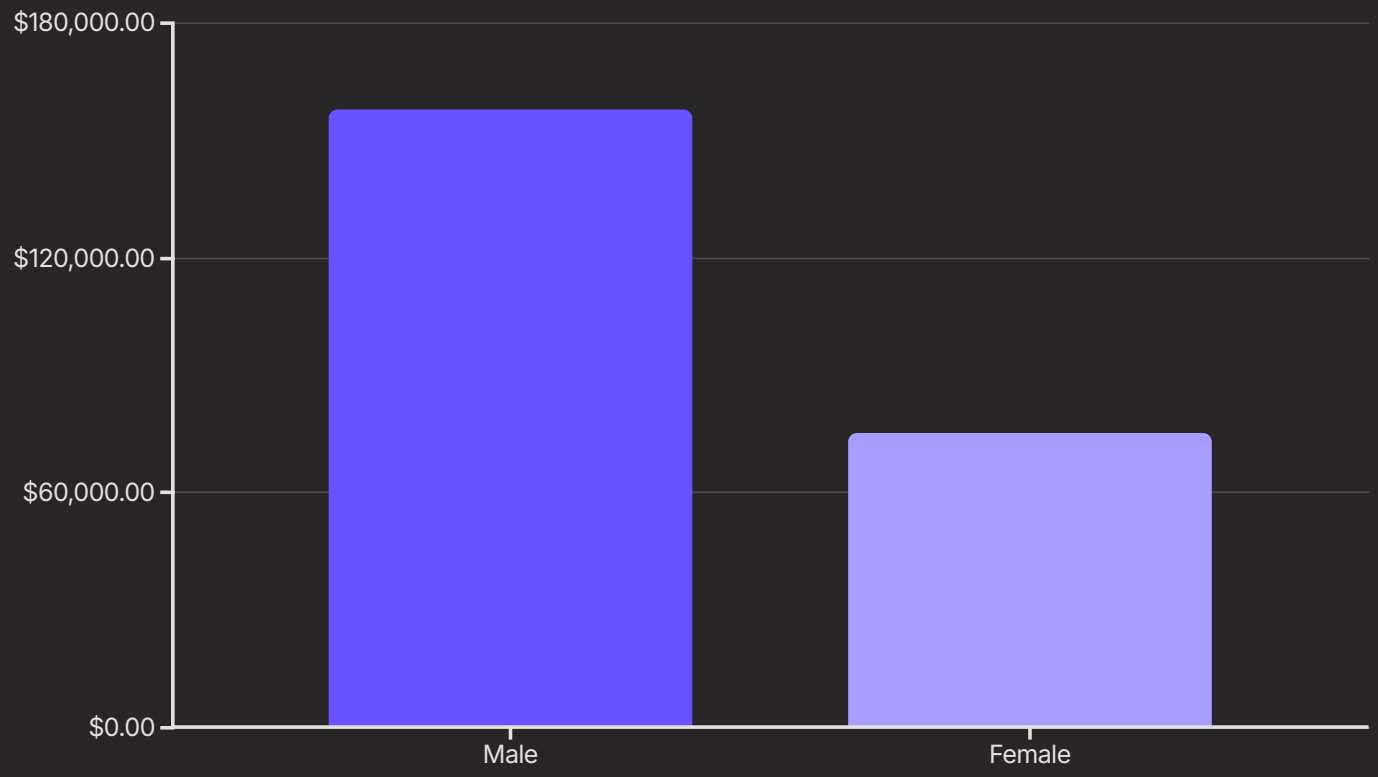
05

Data Reduction

Removed the redundant 'promo_code_used' column due to identical values with 'discount_applied'.

Made with GAMMA

Data Analysis using SQL: Revenue by Gender



SQL analysis reveals a significant difference in revenue contribution between genders.

Male customers generated $157,890 in total revenue, while female customers contributed $75,191.

This insight can inform targeted marketing strategies.

Made with GAMMA



SQL Insights: Top Products & Shipping

Top 5 Products by Rating

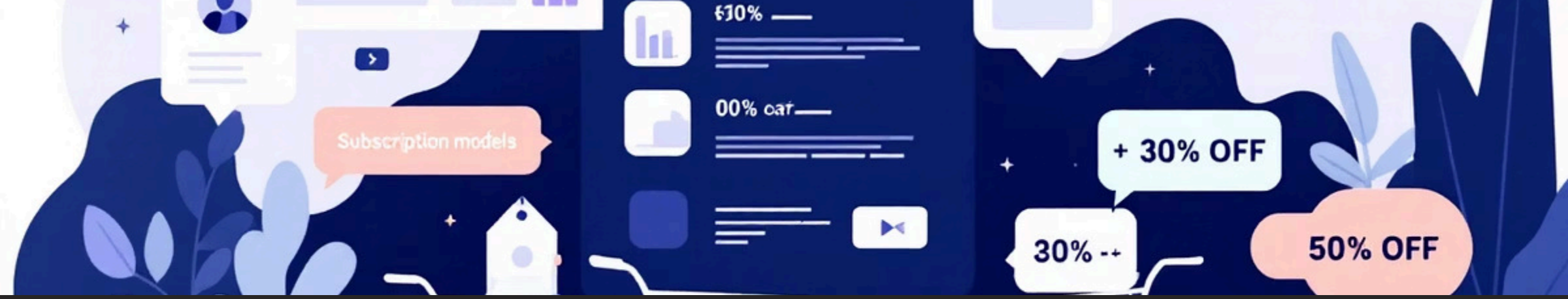
- Gloves: 3.86
- Sandals: 3.84
- Boots: 3.82
- Hat: 3.8
- Skirt: 3.78

Average Purchase by Shipping Type

- Express: $60.48
- Standard: $58.46

Customers using Express shipping tend to have slightly higher average purchase amounts.

Made with GAMMA



SQL Insights: Subscription & Discounts

Subscriber vs. Non-subscriber Spend

- Subscribers: 1053 customers, Avg Spend $59.49, Total Revenue $62,645
- Non-subscribers: 2847 customers, Avg Spend $59.87, Total Revenue $170,436

Non-subscribers contribute more to total revenue due to higher numbers, despite similar average spend.

Top 5 Products with Discount

- Hat: 50.00%
- Sneakers: 49.66%
- Coat: 49.07%
- Sweater: 48.17%
- Pants: 47.37%

These products frequently benefit from discounts, influencing purchase decisions.

Made with GAMMA

Customer Segmentation & Repeat Buyers

Customer Segments

- Loyal: 3116 customers
- Returning: 701 customers
- New: 83 customers

The majority of customers are loyal, indicating strong retention.

Repeat Buyers & Subscription

- Repeat Buyers (5+ purchases) who Subscribe: 958
- Repeat Buyers (5+ purchases) who Don't Subscribe: 2518

A significant portion of repeat buyers are not subscribed, presenting an opportunity for conversion.



Made with GAMMA



Top Products by Category & Age Group Revenue

Top 3 Products in Each Category

- **Accessories:** Jewelry, Sunglasses, Belt
- **Clothing:** Blouse, Pants, Shirt
- **Footwear:** Sandals, Shoes, Sneakers
- **Outerwear:** Jacket, Coat

Revenue by Age Group

- Young Adult: $62,143
- Middle Aged: $59,197
- Adult: $55,978
- Senior: $55,763

Young Adults contribute the highest revenue, followed closely by Middle Aged customers.

Made with GAMMA

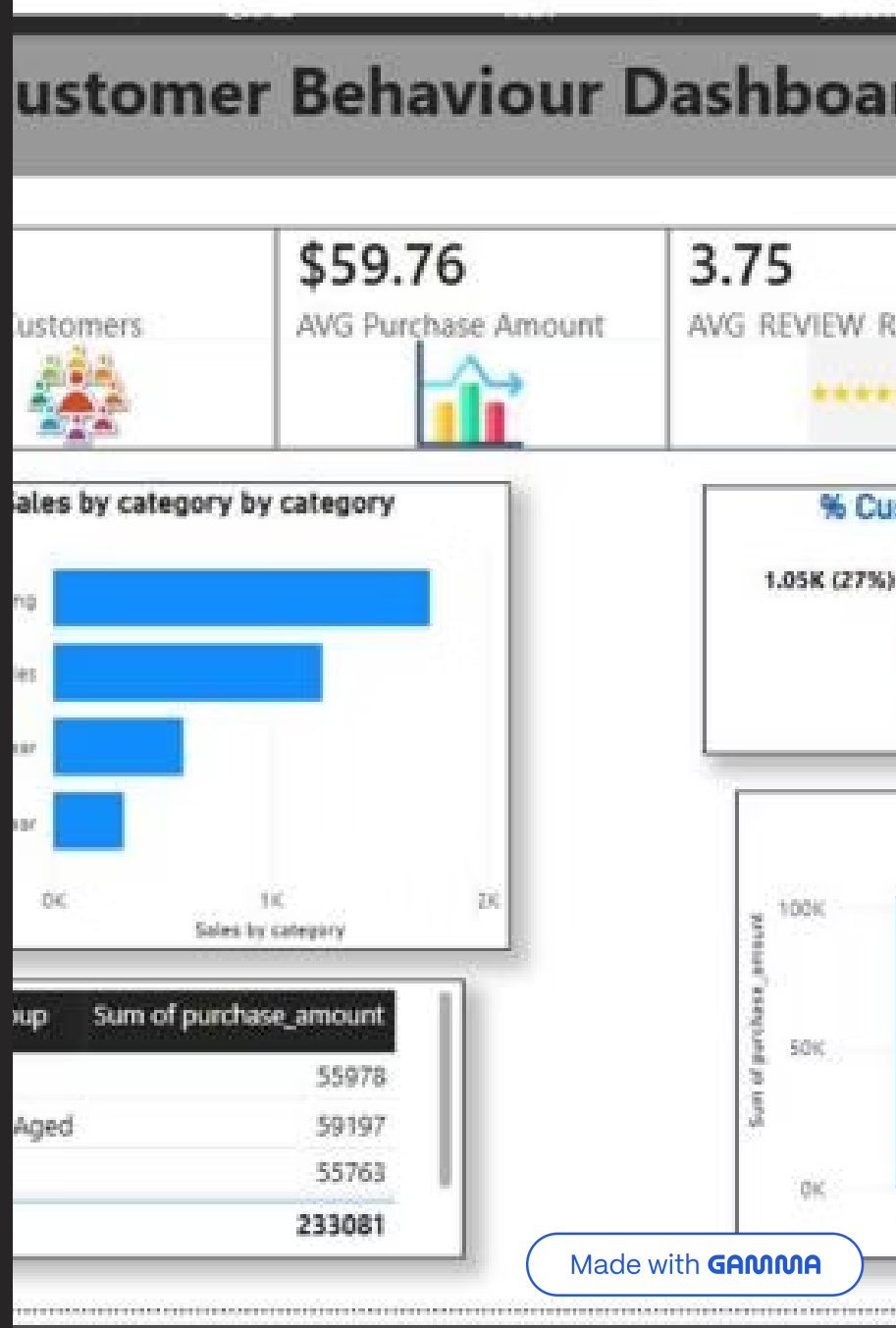
Customer Behavior Dashboard

The PowerBI dashboard provides a comprehensive visual summary of customer behavior, including sales by category, subscription status, and overall metrics.

4K
No. of Customers

$59.76
AVG Purchase Amount

3.75
AVG REVIEW RATING



Business Recommendations

Boost Subscription

Promote exclusive benefits to increase subscriber numbers.

Customer Loyalty Programs

Implement rewards for repeat buyers to foster loyalty.

Review Discount Policy

Balance sales boosts with careful margin control.

Product Positioning

Highlight top-rated and best-selling products in marketing campaigns.

Targeted Marketing

Focus efforts on high-revenue age groups and express-shipping users.

Made with GAMMA